GEOptimize

167 Lombard Avenue, Suite 500
Winnipeg, MB R3B 0V3
+1 (204) 318-2156

Proposal for:	Proposal Date:	Proposal Number:
CHA 3 Winners Circle, Suite 100 Albany, NY 12205-0269 ATTN: Brendan Hall, PE, BEMP	2024-01-16	QU-0133

Reference
US NY CHA Troy District

Understanding of Project

National Grid, in collaboration with the Troy Local Development Corporation ("Troy LDC"), has identified a site studied in NYSERDA PON 4614. The proposed pilot site is an urban, densely populated area, and is bordered by Front Street, Fulton Street, and Broadway in downtown Troy. The proposed thermal energy resource is a geothermal bore field in Riverfront Park, with all thermal energy generation equipment and associated piping to be installed, owned, operated, and maintained by Troy LDC. The total connected load for the Troy UTEN Pilot that the bore field can support is estimated at 550 tons, serving 350,000 square feet. CHA has completed a district geothermal design for this project and has requested TRNSYS modelling from GEOptimize for their design.

For the purpose of this proposal, GEOptimize has assumed 6 separate buildings will be modelled. GEOptimize has also assumed CHA will provide individual hourly loads for each of the 6 separate buildings.

District Geothermal System Modelling

To model and design the layout of the GSHP district energy system, TRNSYS V18.03 will be used. TRNSYS is a simulation program used to model large, interconnected energy systems. Energy loads for each individual building provided by CHA will be imported into the district energy model to simulate the building loads and their impact on the district energy system. To obtain a starting point on the number of boreholes needed for the aggregate district load, GLD software will be used. Once a rough number of boreholes is determined, TRNSYS software is then used to analyse the relationship between the GHX's, buildings, and other potential energy resources. Other energy sources and / or heat sinks that can be modelled with TRNSYS are wastewater heat transfer, heat dissipation with a snow melt system, heat injection from solar thermal collectors, open water well heat exchangers, pond heat exchangers, heat dissipation from fluid coolers or cooling towers, etc. A series of simulations of the model are developed to optimize system performance including:

- The size and configuration of GHX modules
- Location of GHX modules in relation to building heating and cooling requirements

- System control strategy

Information Required

- 8760 hourly energy models for each of the 6 separate buildings
- GHX detailed drawings
- Thermal conductivity test report or geotechnical report for soil thermal properties
- Pump and pipe specifications

Scope of Work

1) Perform two iterations of a TRNSYS model for the district system
2) TRNSYS modelling knowledge sharing sessions with CHA
3) Summarize inputs/outputs from TRNSYS modelling iterations in PowerPoint document including:
    a. Heat exchange fluid temperature along various points on the ambient loop
    b. Building inlet/outlet temperatures
    c. GHX inlet/outlet temperatures
    d. Ambient loop flow rates
    e. Ambient loop pumping control strategy
    f. Annual pump power consumption

Deliverables

GEOptimize will prepare a PowerPoint presentation to be delivered to CHA at a meeting via screen sharing software detailing 2 iterations of the district system design and their respective inputs/outputs.

Pricing

Description			Line Total
TRNSYS Modelling			$ 15,900.00
Total (USD)			**$ 15,900.00**

Terms

- Additional out of work scope will be charged at 250 USD/hr.
- Work will not commence until a signed proposal and purchase order number is issued.

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